

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1st Year 1st Term Regular Examination, 2021
HUM 1117
(Functional English)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Make sentences on the following structures using the verb given in bracket (14)
 - i) Subject + Intransitive verb + adverbial (run as verb)
 - ii) Subject + Linking verb + adjective complement (grow as verb)
 - iii) Subject + Linking verb + noun complement (look as verb)
 - iv) Subject + Transitive verb + obj. + noun complement (elect as verb)
 - v) Subject + Transitive verb + obj. + adj. complement (prove as verb)
 - vi) Subject + Transitive verb + obj. + obj. (make as verb)
 - vii) There + verb + subject + extension (any verb)
- (b) Change the following words as directed and make sentences with the changed words. (12)
People (into adjective); People (into verb); People (into noun); Depend (into adjective); Depend (into noun); Free (into verb)
- (c) Make a new word with the following prefixes and suffixes and use the new words in sentence. (09)
Auto....., By....., Co.....,ing,scape,ance
2. (a) Transform the following sentences as directed. (14)
 - i) Sheraton is the most expensive restaurant. (positive)
 - ii) I reached the station and the train started. (complex)
 - iii) When I woke up, it was dark. (compound)
 - iv) Let us help the poor. (assertive)
 - v) It could be a child again. (assertive)
 - vi) There is nothing wrong with me. (interrogative)
 - vii) Only a cheat can do it. (negative)
- (b) Make use of the following words in sentence as asked in brackets. (12)
Pen (as verb), Watch (as noun), Brick (as verb), About (as adverb), About (as adjective), Above (as adjective)
- (c) Supply a suitable word to fill in the blanks. (09)
 - i) What you do if he failed to turn up.
 - ii) Their efforts resulted into a
 - iii) the fire, we ran away in fear.
 - iv) He said he go there the next day.
 - v) He is in speaking English.
 - vi) She went to a doctor for
3. (a) Make wh question with the underlined words of the following sentences. (14)
 - i) He has been preparing for the examination for 3 hours.
 - ii) Liza reads the book with a great eagerness.
 - iii) Biva is five feet and three inches tall.
 - iv) Mathematics is a difficult subject.
 - v) The car has a top speed of 80 miles per hour.
 - vi) He is trying hard to do a good result.
 - vii) He took a few deep breaths before death.
- (b) Make sentences using the following modals as directed. (12)
 - i) May (To express guess about the future)
 - ii) Must (To express inference)
 - iii) Could (To express polite request)
 - iv) Need (To express unnecessary actions in the past)
 - v) Had better (To express preference)
 - vi) Should ((To express duty in the past)

- (c) Write one antonym and one synonym for the words given below and make sentence with the antonyms and synonyms. (09)
Tension, Love, Sincere
4. (a) Make sentences expressing the following emotions/notions. (14)
i) Approval, ii) Disapproval, iii) Proposal, iv) Good wish, v) Regret, vi) Introduction, vii) Sympathy
- (b) Complete the sentences with sub ordinate clauses. (12)
i) I wonder (noun clauses)
ii) The man was far away. (adjective clauses)
iii) I will let you know (adverb clauses of time)
iv) He resigned the post (adverb clauses of purpose)
v), it vaporizes. (adverb clauses of condition)
vi) He speaks (adverb clauses of manner)
- (c) Make sentence with the following idioms and phrases. (09)
Palmy day, Foregone conclusion, Bad egg, Die in harness, cry down, lay waste

Section – B

5. (a) Read the following passage carefully and answer the questions that follow. (20)
- Unfair practices instead studying sincerely, that take place of suicide for a student, can destroy the life of a student and consequentially a country's future. Here a student doesn't grow up with knowledge from the reading material. The student may think that not studying properly, only studying notes created by other ones of his/her society he/she may do well in life. But this philosophy corrodes one's life and this evil idea brings our nation in darkness. They should remember the life of enlightenment, which gives birth humans a substantial structure of a human. Here lie the distinct differences between a gentleman and ungentle man, a responsible man and an irresponsible man, an egocentric man and an altruistic man, above all an expected human and a burden human for the world, etc. The young man, so, in the mending wall' by Robert Frost reflects the life of cognition that as a humans are not beast, they (he and the old man) need not mend the crack of the wall: 'But here they are no cows: if there would be cows, a symbol of ignorance, there would need repair for the wall. How is it possible to become a human? Sincerely studying more and more can create an enlightened man. Likewise, other evil ways, we call these ways evil logically, such as not maintaining the style of life for the sake of a sound and safe health all the day long or the majority of the time being set for the whole of a semester, are decided to lead an unhealthy life. This can in the long run exist a deceased body. A life of mental and physical progression can only beautify basically the life of a nation, a nation hopes from their students.
- i) What are the unfair practices of a student? What do they do?
ii) Why are they though as destructive?
iii) What does an enlightened man do? And how is it?
iv) How can a student grow a healthy life?
- (b) Make a precis of the above written passage [Q: 5(a)] with a suitable title. (15)
6. (a) Amplify the idea given below. (20)
How poor are they that have not patience! What would did ever heal but by degrees?
- (b) Write a paragraph on perseverance. (15)
7. (a) Write a report on your Language Lab. (20)
(b) Prepare a CV along with a job application. (15)
8. Write a free composition on any one of the following. (35)
i) Quality Education: Good Governance
ii) Corruption: the main problem in Bangladesh

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Ch 1117
Environmental Chemistry

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Differentiate between macromolecules and polymers. (08)
(b) Write the mechanism of free radical polymerization of polyacrylonitrile. (11)
(c) Whether polyacetylene is a conducting polymer or not? If it is a conducting polymer explain the conducting mechanism. (10)
(d) Why ionic polymer is called living polymer? (06)
2. (a) What is meant by inversion of configuration? "AN S_N2 reaction proceeds with complete Stereochemical inversion"- Justify the statement. (09)
(b) Define rate-determining Step. Discuss in detail the S_N2 versus S_N1 reaction. (10)
(c) Why 1° Carbocation are less stable than 2° or 3° Carbocation? (08)
(d) Racemization takes place on which type of nucleophilic substitution reaction? Explain with a suitable example. (08)
3. (a) What is environmental chemistry? Briefly describe the segments of the environment. (12)
(b) How would you broadly divide the major regions of the atmosphere? State their respective altitudes and temperature ranges. What are the important chemical species in each region? (10)
(c) How to create photochemical smog? - Explain with diagram. (13)
4. (a) Define Oligotrophic and Eutrophic lake. (08)
(b) Discuss the domestic wastewater treatment process with flow diagram. (12)
(c) Is ozone layer very much useful for the living beings? Explain why? (07)
(d) Do humidity and dust particle have any effect on the rate of corrosion? Explain. (08)

Section – B

5. (a) Why alkali metals are not found in the free state in nature? Explain with some reactions. (06)
(b) Compare the properties of alkali and alkaline earth metals. (12)
(c) Halides are weak ligands compared to carbon based ligands. Why? (08)
(d) Complete the following statements for the complex ion, (09)
 $[Cr(C_2O_4)_2(H_2O)_2]^-$:
 - i. The oxidation number and coordination number of the central metal;
 - ii. Identify the bidentate ligand;
 - iii. The effective atomic number of the central metal.
6. (a) Distinguish between double salts and complex salts with suitable examples. (06)

- (b) Compare the magnetic properties and hybridization of $[Fe(CN)_6]^{3-}$ and $[Fe(Cl)_6]^{3-}$ complex ions with the help of valence bond theory. (14)
- (c) Draw the molecular orbital diagrams of O_2 , O_2^{2-} and O_2^- . Compare their bond strength and bond length. (15)
7. (a) Explain the abnormal behavior of H_2O molecules with definite cause. (08)
- (b) Define colligative properties of dilute solutions and write down the different names of colligative properties. (06)
- (c) Why conjugate base of a strong acid is weak on the other hand conjugate base of a weak acid is strong? Explain with suitable examples. (12)
- (d) Define the following terms: i. Dipole-dipole interaction; ii. London-force and iii. Lewis acid and Lewis base (09)
8. (a) How do nuclear reactions differ from ordinary chemical reactions? (08)
- (b) Complete the following nuclear equations and identify X in each case: (08)
- $^{135}_{53}I \rightarrow ^{135}_{54}Xe + X$
 - $^{40}_{20}K \rightarrow ^0_{-1}\beta + X$
 - $^{59}_{27}Co + ^1_0n \rightarrow ^{56}_{25}Mn + X$
 - $^{235}_{92}U + ^1_0n \rightarrow ^{99}_{40}Zn + ^{135}_{52}Te + 2X$
- (c) Half-life period of Thorium is 24.5 minutes. How much Thorium would be left after 30 minutes if the initial amount of thorium is 1.0 gram? (10)
- (d) Discuss the stability of nucleus in terms of neutron-proton ratio. (09)

Khulna University of Engineering & Technology
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BURP 1stYear 1st Term Examination, 2021

Math 1117
(Mathematics-I)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Figures in the right margin indicate full marks.
ii) Assume reasonable value for any missing data.
iii) Answer any three questions from each section in separate script.

Section – A

- 1 (a) What is meant by invariants? Find the transformed equation of $4xy - 3y^2 - 5 = 0$ when the axes are rotated through an angle $\tan^{-1}(2)$. 13
- (b) Find the rotational angle of the axes to remove the xy -term from the equation $3x^2 + 2\sqrt{3}xy + 3y^2 - 1 = 0$ 06
- (c) Reduce the equation $x^2 - 4xy + y^2 + 8x + 2y - 5 = 0$ to the standard form. Find also the equation of latus rectum and axes. 16
- 2 (a) Define the following functions with example for each : 12
- (i) Explicit function
- (ii) Implicit function
- (iii) Logistic function
- (b) Sketch the graphs of the following functions with domain and range: 12
- (i) $f(x) = \sinh x$
- (ii) $g(x) = \cosh x$
- (iii) $h(x) = \operatorname{sech} x$
- (c) what do you mean by echelon form of a matrix? Reduce a matrix, 11
- $$A = \begin{pmatrix} 1 & 2 & -2 \\ 2 & 5 & -4 \\ -1 & -3 & 2 \\ 2 & 4 & -1 \end{pmatrix}$$
- into echelon form
- 3 (a) explain the following terms 12
- (i) Idempotent matrix
- (ii) Hermitian matrix
- (iii) Skew-symmetric matrix.
- (b) Find the symmetric and skew-symmetric parts of a matrix, 11
- $$A = \begin{bmatrix} 1 & 2 & 4 \\ 6 & 8 & 1 \\ 3 & 5 & 7 \end{bmatrix}$$
- (c) If exist, then find the inverse of a matrix, 12
- $$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$$
- By using elementary transformations.
- 4 (a) Determine the values of k so that the following systems in x, y, z has (i) a unique solution (ii) an infinite number of solutions where, 14
- $$\begin{aligned} x + y - z &= 1 \\ 2x + 3y + kz &= 3 \\ x + ky + 3z &= 2 \end{aligned}$$
- (b) Solve the following system linear equations with the help of matrices: 13
- $$\left. \begin{aligned} x + 2y - 3z &= 6 \\ 2x - y + 4z &= 2 \\ 4x + 3y - 2z &= 14 \end{aligned} \right\}$$
- (c) Discuss about singular and non-singular matrices with examples. 8

Section – B

- 5 (a) Find the rectangular and spherical coordinates of a point, whose cylindrical coordinates are $(3, \frac{2\pi}{3}, 4)$ 12
- (b) Define direction cosine of a line. if l, m, n are Dc's of a line, then prove that $l^2 + m^2 + n^2 = 1$. Also find the Dc's of the lines which lies on the xy -plane and makes equal angles with the x -axis and y -axis. 11
- (b) Find the angle between any two diagonals of a cube. 12
- 6 (a) Determine the length of projection of a segment line joining the points $A(1,2,3)$ and $B(5,-2,1)$ on the x -axis. Also find the angle between the AB line and the y -axis. 12
- (b) Find the equation of the plane through the points $(2,2,1)$ and $(9,3,6)$ and perpendicular to the plane $2x + 6y + 6z = 9$ 11
- (c) Find the distance from the point $(1, -2, 3)$ to the plane $x - y + z + 1 = 0$ measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z-1}{-6}$ 12
- 7 (a) Prove that the two lines $x = a_1y + b_1, z = c_1y + d_1$ and $x = a_2y + b_2, z = c_2y + d_2$ are perpendicular to each other if $a_1a_2 + c_1c_2 + 1 = 0$ 10
- (b) Test whether the four points $(1,1,1), (2,1,-1), (3,3,0)$ and $(0,-1,0)$ are coplanar or not. 11
- (c) Show that the lines $\frac{x+1}{2} = \frac{y-1}{2} = z$ and $\frac{x-1}{6} = \frac{z-3}{5} = y+1$ are coplanar. Also find their point of intersection and the equation of the plane containing them. 14
- 8 (a) What do you mean by spherical triangle? Establish the following relation among the sides and angle of a spherical triangle ABC :

$$\cos c_s = \cos a_s \cos b_s + \sin a_s \sin b_s \cos C$$
where a_s, b_s and c_s represent the sides and C represents the angle of the spherical triangle ABC 13
- (b) Find the distance between Dhaka ($23^\circ 44'N, 0.24^\circ E$) and San Francisco ($37^\circ 44'N, 122^\circ 27'W$) in nautical miles. 10
- (c) Determine the great circle distance between two cities, Dhaka ($23^\circ 44'N, 0.24^\circ E$) and Tokyo ($35^\circ 40'N, 139^\circ 45'E$) in nautical miles. 12

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BURP 1st Year 1st Term Examination, 2021.
URP 1111
(History/Evolution of Human Settlement)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Figures in the right margin indicate full marks.
iii) Answer any **three** questions from each section in separate script.

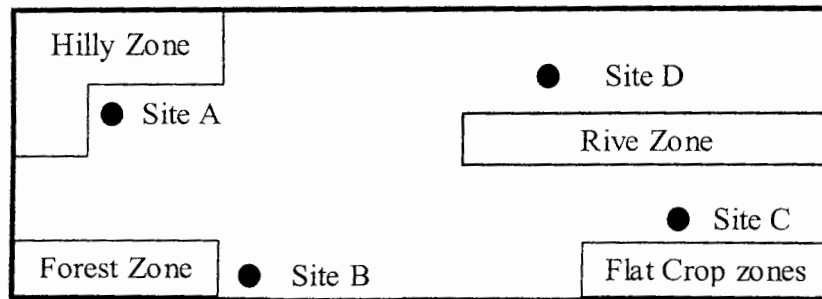
Section – A

1. (a) Mention the major innovations or evolutions that occurs during: (15)
(i) Stone Age; (ii) Bronze Age; (iii) Iron Age.
(b) Write short notes on the following: (10)
i) Beam Frame house
ii) Sedentary lifestyle.
(c) Mention the various stages in the evolution of human settlement. List out the site factor of human settlement. (10)
2. (a) Classify towns based on their functional classification and rural settlements on patters with proper diagrams. (15)
(b) Compare between Urban and Rural settlement characteristics. (10)
(c) Describe with examples, the bases of classifying a settlement as urban around the world. (10)
3. (a) Briefly discuss the concept of the Three Magnet Theory. Draw a “pizza slice” of the Garden City with proper labels depicting the layout. (15)
(b) Describe the Vertical City of Le Corbusier in terms of its street system and population distinction. (10)
(c) Write about real world examples of Garden city, Linear City and Vertical City. (10)
4. (a) Differentiate between the following: (10)
i) Primitive Non-organized and Primitive Organized.
ii) Static Urban and Dynamic Urban Settlement.
(b) How did fortification develop during Iron age? (10)
(c) Write short notes w.r.t. human settlements for: (15)
i) Pundravardhan; ii) Harikela; iii) Bagerhat.

Section – B

5. (a) What do you mean by content and container in human settlement? (05)
(b) “Sites provide better situation for settlement” – Describe the statement with examples. (10)
(c) Describe the major city features of the Mesopotamian Civilization. Also, describe the characteristics of city planning concept in Mesopotamian Civilization. (20)

6. (a) Define human settlement in term of geospatial modeling. (05)
- (b) Illustrate the concept 'Ecumenopolis'. From the following figure you have to choose one site among A, B, C and D by applying the scoring method of site selection. (15)



- (c) Why did Mesopotamian Civilization grow? Mention some major features of Mesopotamian Civilization. (15)
7. (a) How were the city states Babylon ruled by its kings? Describe with major contribution. (15)
- (b) Why did Egyptian Civilization grow? Represent the cause and effect of the Egyptian Civilization. (15)
- (c) What do you mean by Set Back Rule? (05)
8. (a) How the 'Citadel' worked in Indus Valley Civilization? Mention some common features of Indus Valley housing infrastructure. (15)
- (b) Represent the features of Greek Parthenon with plan and frost elevation of it. (10)
- (c) What do you mean by Greek Stoa? Mention some features of Hellenistic city. (10)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 1stYear 1st Term Examination, 2021
URP 1113
(Fundamentals of Planning Process)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What is planning? How the term "Planning" comes from the history? Draw the timeline. (15)
(b) Please provide the logical drawbacks of "Comprehensive Planning" policies and its major implication in real world. (15)
(c) Mention the steps of Radical Planning. (05)
2. (a) Suppose, you have multiple options to go to Dhaka from Khulna by Bus/Air/Train. Recall the problem with one of the planning theories and solve the problem. (20)
(b) Draw a comparison matrix between Incremental Planning and Transactive Planning. (10)
(c) Mention the steps of Advocacy Planning. (05)
3. (a) Mention the policies of sustainability pillars in planning and give an appropriate sketch regarding the sustainability. (10)
(b) Planners are the Negotiators; do you agree with the statement? Why or why not? (10)
(c) What is SITAR model. Briefly discuss about the model. (15)
4. Briefly discuss about the followings with practical examples. (35)
a) Bottom-up approach b) Paradigm shift in planning c) Spatial planning d) Inclusive city e) Limitations of Rational Planning.

Section – B

5. (a) What is urban sprawl? Discuss the end result of urban sprawl in a metropolitan city with necessary examples. (10)
(b) What is primate city? Write down the advantages and disadvantages of primate cities. (10)
(c) Briefly narrate the necessity of urban planning with example in context of Bangladesh. (15)
6. (a) Briefly explain the terms Project, Vision, Mission, Goal and Objective with relevant examples especially in context of Urban and Regional Planning. (20)
(b) Differentiate Structure Plan and Master Plan with appropriate examples. (15)
7. (a) Briefly describe the major functions of Planning Commission. (15)
(b) Planning Commission has a three-fold role in socio-economic development of the country. Explain with examples. (10)
(c) "There are different types of institutional linkages of Planning Commission with other Ministries/Divisions/Organizations". Write down the linkages. (10)
8. (a) Suppose you are working as an Urban Planner in Khulna City Corporation (KCC). KCC needs to develop a Solid Waste Management (SWM) Plan for Khulna city. Discuss the stages of its conventional planning process. (20)
(b) Draw the figure "A Ladder of Citizen Participation" and explain its stages with relevant examples. (15)